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A Living Lab approach to needs-driven policy design and policy implementation through interconnected Double Diamond processes

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Abstract

This article explores how Living Lab principles and a Double Diamond framework support needs-driven rural digital policy design and context-sensitive policy implementation in cross-border contexts. The study is situated in the Torne Valley, a Finnish Swedish rural border region characterised by fragmented governance structures, long distances, and limited institutional coordination. Building on previous work on Living Lab-inspired policy co-creation, the article investigates how knowledge generated through participatory co-creation can be translated into institutional practice. Following Design Science Research Methodology, the study develops a policy model structured around two interconnected Double Diamond processes. In the first process, Living Lab principles function as generative mechanisms that enable participatory co-creation, generating an understanding of rural needs and local contexts that is structured into needs-driven knowledge for policy design. In the second process, the Living Lab principles are reapplied within institutional contexts, generating an institutional understanding of rural needs and contexts that creates institutional learning and supports context-sensitive policy implementation. Empirical material was collected through participatory workshops, interviews, focus groups, and collaborative Living Lab activities across four policy areas. The article contributes to Living Lab and policy implementation research by demonstrating how participatory knowledge can be translated into institutional learning through interconnected Double Diamond processes.

Key words

Living Labs principles, policy implementation, needs driven policy, Double Diamond framework, cross-border governance, generative mechanisms



Introduction

Public organisations possess structural discretion and professional knowledge that grant them an autonomous capacity for action (Howlett, 2019). At the same time, digital transformation (DT) has become a central component of contemporary governance, shaping how public services are organised and delivered (Lindberg, 2024). While national and European strategies for rural digitalisation exist, they are frequently implemented through top-down mechanisms and shaped by established notions of industrial modernity (Nyhlén & Gidlund, 2022). These approaches tend to overlook the complexity of place-based realities and the diversity of rural experiences (Pfaffl, 2019). This points to a broader implementation gap, where formally designed policies fail to translate into context-sensitive practices that reflect local needs and contexts. (Pekola, 2024).

This article focuses on the Torne Valley (Tornionjokilaakso/Tornedalen), a Finnish Swedish border region characterised by shared social, cultural, environmental, and economic conditions, but lacking integrated policy (Pekola, 2024). Rural areas are often marginalised due to infrastructural limitations, demographic decline, and limited institutional capacity (Roberts et al., 2017), as well as insufficient understanding of local conditions (Pfaffl, 2019). In cross-border contexts, governance fragmentation and institutional misalignment further constrain coordination (Hassink & Lagendijk, 2001; Karagkouni & Dimitriou, 2025), meaning that formal discretion does not necessarily enable action aligned with local needs and local contexts (Habibipour et al., 2026; Lindberg et al., 2024; Meier & Bothe, 2001).

In practice, this misalignment gives rise to informal and adaptive modes of governance. It can take the form of guerrilla government, in which officials informally express dissent and attempt to influence policy outcomes based on their own interpretations of what is right or necessary (O'Leary, 2019), as well as street-level bureaucracy, where frontline actors interpret and enact policy under constrained conditions (Lipsky, 2010). These dynamics highlight how implementation unfolds beyond formal structures, relying on situated judgement and improvisation. For such practices to be sustained and made productive, organisations must rely on the practical judgement of officials, enabling context-sensitive action within institutional constraints (Bornemark, 2020). This becomes increasingly important in governance contexts characterised by complexity and interdependence. As Nyhlén (2013, p. 1) notes, "The politics is increasingly characterised by project and process politics, networks, cooperation and partnerships". Governance for DT can therefore be understood as a socio-technical and inherently ambiguous process, where implementation depends on how public organisations interpret, negotiate, and adapt policy to local contexts rather than simply applying formal rules (Väyrynen et al., 2025). Ambiguity shifts decision-making from regulation itself to the actors who interpret and apply it, requiring continuous adjustment in response to evolving conditions (Khatunov, 2024). In rural contexts, effective implementation therefore depends on developing a shared understanding of local needs and conditions, enabling public organisations to translate formal policy into context-sensitive practice (Lindberg et al., 2024).

In response, a growing body of research advocates for more inclusive and participatory approaches to policymaking that are sensitive to local contexts (Bergvall-Kåreborn & Ståhlbröst, 2008; Cognetti & Maranghi, 2023; Habibipour et al., 2021). Living Labs (LL) have emerged as a promising participatory approach to policymaking. In this article, we draw on LL principles to support participatory co-creation in real-life settings (Bergvall-Kåreborn, 2009; Galvagno & Dalli, 2014; Schuurman, 2015; Ståhlbröst & Holst, 2017). While LL principles have been widely applied in service



and product innovation, their role in policy development remains underexplored, particularly in rural and cross-border contexts (Veeckman & Temmerman, 2021; Rizzo et al., 2021). Through participatory co-creation, these principles enable participants to develop a shared understanding of rural needs and local contexts, providing the foundation for policy design. Building on these participatory principles, the Double Diamond (DD) framework provides a structured process for structuring knowledge generated through co-creation into policy design (Kochanowska et al., 2022), yet limited attention has been given to how such approaches support policy implementation and institutional adaptation. A previous study showed how LL principles can be integrated with the DD framework to support structured digital policy co-creation, resulting in early-stage policy outputs and practical facilitation materials (Habibipour et al., 2026). However, limited attention has been given to how the knowledge generated through this structured co-creation process supports policy implementation and institutional adaptation.

The key gap therefore lies in understanding how knowledge generated through participatory co-creation can be translated into institutional understanding and subsequently embedded in policy implementation. Building on the premise that governance and practice are mutually constitutive (Väyrynen et al., 2025), this article conceptualises institutional learning as the mechanism linking knowledge generated through needs-driven co-creation with formal policy implementation.

The study draws on empirical work conducted in the Torne Valley, combining community-based data collection with interdisciplinary collaboration and practitioner engagement within a LL approach.

The study is guided by the following research questions:

RQ1: How can Living Lab principles generate knowledge throughout the needs-driven and institutional policy processes?

RQ2: How does a Double Diamond/ Living Labs framework structure and influence institutional learning during the implementation of rural digital policy?

By addressing these questions, the article contributes to the literature on governance for DT and rural innovation by advancing understanding of how LL principles, applied through two interconnected DD framework processes, generate an understanding of rural needs and local contexts through participatory co-creation, structure this understanding into policy design, and subsequently support institutional learning for context-sensitive policy implementation.

Previous research

This article focuses on the shift from technologically driven LLs to socially oriented LLs, including both urban and rural LLs, that emphasise empowerment, inclusivity, local leadership, and capacity-building (Cognetti & Maranghi, 2023; Chronéer et al., 2019). In rural contexts, these principles also contribute to enhanced digital services and stronger rural retail, with the overall aim of improving quality of life in villages (Habibipour et al., 2021; Lindberg, 2024).

ULLs have the potential to influence policy by making local, often overlooked, competencies visible. This shift helps transform how residents are perceived; from being passive “targets” of interventions to active “agents” in policymaking (Cognetti & Maranghi, 2023). A similar dynamic is evident in rural development, where women often engage through unofficial networks beyond formal organizations, challenging and reshaping traditional boundaries between public, and private roles (Arora-Jonsson, 2017). This form of grassroots participation is often described as self-organization, where citizens



take the initiative to drive change in their communities (Edelenbos & van Meerkerk, 2016). In a rural LLs in northern Sweden, the implementation of digital services has further supported this by opening new spaces for participation and promoting a relational view of competence (Lindberg, 2024). By engaging in small-scale, robust LLs, such as the SoHoLab and Digiby projects, LLs have provided evidence-based micro-interventions that can inform broader policy agendas (Cognetti & Maranghi, 2023; Lindberg et al., 2025). These studies demonstrate how participatory LL approaches create the conditions for making local knowledge, competencies, and everyday experiences visible, thereby providing a foundation for more context-sensitive policymaking.

From a policymaker's perspective, it is important to create opportunities for direct interaction with residents. One effective method is to establish local "pop-up venues" in villages, which can help reach individuals outside established networks (Lindberg et al., 2024). These interactions should follow a LL approach, prioritizing listening and co-exploration over predefined solutions. The goal is to act as a facilitator; supporting residents and organizations in voicing their perspectives and contributing meaningfully to policy development (Lindberg, 2024). Facilitative practices such as these, create opportunities for policymakers and communities to jointly explore rural needs and develop a shared understanding of local contexts as a foundation for policy development.

This orientation toward facilitation and situated interaction is also reflected in recent work on rural cross-border digital policy co-creation, where LL principles were integrated into the DD framework to structure the Discover, Define, Develop, and Deliver phases of policy work (Habibipour et al., 2026). This work provides a foundation for understanding how knowledge generated through participatory co-creation can be structured into policy design, while leaving open the question of how such knowledge becomes embedded in institutional practice.

Policy learning

Policy learning provides the theoretical perspective for understanding how policy knowledge developed through participatory co-creation and structured through the DD framework is interpreted, transformed into institutional understanding, and embedded within organisational practice. Hall (1993) conceptualises social learning as a process through which policymakers revise their understanding of problems and appropriate responses based on past experience and new information. Rather than viewing policymaking solely as a matter of control or strategic action, he emphasises its interpretive and adaptive nature, describing policymakers as engaged in "collective puzzlement on society's behalf" (Hall, 1993, p. 276). Policy development is therefore shaped by experience-based reflection, as actors continuously reassess what works, what does not, and why. Learning is embedded in the policy process itself and is typically gradual, characterised by ongoing interpretation and adaptation, although more substantial shifts may occur when existing understandings lose credibility (Baumgartner, 2013). Moreover, learning processes are shaped by interactions between policy instruments and underlying ideas, where feedback informs how actors interpret and adjust their approaches (Daugbjerg & Key, 2020). In the context of this study, these processes extend beyond individual policymakers to the institutional level, where policy knowledge developed through participatory co-creation is collectively interpreted to develop an institutional understanding of rural needs and contexts, providing the basis for institutional learning.

To fully understand rural policy contexts, social learning is complemented by causal and diagnostic learning. Causal and diagnostic learning refers to the process through which actors refine their



understanding of cause-and-effect relationships and improve their diagnosis of policy problems (Cerna, 2013; Levy, 1994). It emphasises identifying underlying drivers and assessing which interventions are likely to be effective (Blom & Morén, 2011; Henfridsson & Bygstad, 2013). For example, low uptake of digital services in rural areas may reflect not only infrastructural gaps but also limited value, digital skills, or trust resulting from top-down implementation strategies. In this sense, actors refine both their understanding of causal relationships (Levy, 1994) and the situations they confront (Bergvall-Kåreborn, 2009). Learning is therefore directed not only towards identifying effective interventions but also towards understanding how outcomes are produced through underlying processes (Henfridsson & Bygstad, 2013).

Method

The setting of this article is the cross-border Torne Valley region, focusing on the villages of Muonio and Muodoslompolo. The latter is a small village with a population of around 70 residents in the municipality of Pajala, situated approximately 110 kms north of the municipality's central town. The village hosts a privately owned grocery store and has businesses in tourism, culture, and natural resources. On the Finnish side of the border, Muonio has approximately 2500 inhabitants and offers well-developed community services, including schools, shops, health-care centre and hotels. Muonio's economy is largely oriented toward tourism. In the Muonio Mountain area, concession-based reindeer herding is central to both cultural identity and economic activity. Developed land and natural forests shape the landscape. Torne Valley architecture includes enclosed farmyards reflecting a significant rural heritage. The Torne River catchment waterways support fishing, recreation, and nature restoration, and are part of a Natura 2000 site with high ecological and good chemical status.

Design Science Research Methodology (DSRM)

This study follows DSRM (Peffers et al., 2007) to develop and iteratively refine the artefact: a needs-driven policy model that integrates LL principles into an interconnected DD framework to support institutional learning during policy implementation.

The research was conducted in three iterations, combining empirical data collection, analysis, and iterative testing of the artefact. This approach was selected due to the holistic and multi-domain nature of the policy process, which addresses a wide range of policy areas emerging from local community needs (Roberts et al., 2017). The research involved multiple researchers working across different domains, resulting in an initially fragmented process reflecting the diversity of needs and perspectives (Bergvall-Kåreborn & Ståhlbröst, 2008).

DSRM provided structure to support the integration and alignment of these contributions, enabling a shared understanding of the overall policy process. Through its iterative and design-oriented logic, DSRM facilitated the streamlining of the process and supported collective learning within the research group. It allowed researchers to situate their individual inputs within a common structure, thereby improving coordination and enabling more effective knowledge contribution across policy areas.

In the first cycle, problem identification was conducted through a research workshop, supported by a literature review and thematic analysis of interviews and discussions, resulting in design



objectives for the needs-driven policy model. Following LL principles, stakeholder mapping and empirical grounding were carried out through interviews, informal interactions, school-based sessions, and participatory workshops. Our workshops combined practice-oriented engagement with analytical rigour, enabling participants to contribute within structured yet adaptable objectives (Ørngreen & Levinsen, 2017). The participatory process employed a range of methods, including cultural probes, personas and scenarios, miracle discussions, design fiction workshops, and drama- and activism-based approaches (see e.g., Iivari et al., 2024, 2023; Sebastian et al., 2024; Sharma et al., 2023; Ventä-Olkkonen et al., 2025). This phase was subsequently complemented by demonstration and evaluation activities conducted in collaborative sessions and focus groups (Bannon et al., 2018). Together, these activities established the empirical and participatory foundation of the artefact, resulting in the first needs-driven version of the policy model.

In the second cycle, the artefact was iteratively refined by extending the initial needs-driven policy model into an interconnected DD framework. This second process reapplied LL principles within institutional contexts to translate needs-driven policy knowledge into institutional understanding, thereby supporting policy implementation. The refined artefact was demonstrated in policy contexts as scientific input to Sweden's national rural policy (Lindberg, 2026), as well as in educational settings, and subsequently evaluated with stakeholders (Ståhlbröst & Holst, 2017).

The third cycle focused on evaluating the artefact through the lens of institutional learning. The interconnected DD framework was applied across five cross-border policy domains in the Torne Valley region: education, infrastructure, environment, healthcare, and information sharing (Final Project Report, Cross Border Rural Digital Policy, in writing, 2026). The model was also applied to develop a policy proposal for Sweden's national rural policy (Lindberg, 2026) and evaluated through five student group projects. Together, these applications assessed the artefact's adaptability across different policy contexts and examined how needs-driven policy knowledge developed into institutional understanding and became embedded in policy implementation (Wikström et al., 2024; Howlett, 2019).



Iter	DSRM phase	Activity / Method	Objectives	Target group	No	Output
1st	Problem identification	Start-up research workshop in CBRDP	Identify challenges related to rural digitalisation and cross-border policy implementation	Researchers	1	Initial problem framing and identification of policy challenges
1st	Problem identification	Theoretical background	ibid.	Researcher	1	Litterature overview
1st	Objectives of a Solution	Thematic analysis of workshop and interview data	Derive objectives for a needs-driven policy model based on identified challenges	Researcher		Defined solution objectives for the policy model
1st	Design & Development	Initial framing: Start-up research workshop in CBRDP	Identify key challenges and explore suitable design approaches for rural digital policy	Researcher		Stakeholder mapping, identified challenges, and selection of Double Diamond and Living Labs approach
1st	Design & Development	Empirical grounding: Interviews, school sessions, student case studies, and informal interviews	Generate empirical understanding of local needs and existing practices in rural and cross-border contexts.	Local stakeholders		Empirical data on local needs and collaboration practices
1st	Design & Development	Analysis: Thematic analysis of interviews and focus groups	Synthesise empirical data into design-relevant insights	Researchers		Thematically structured insights informing the design of the policy model
1st	Demonstration	Research workshop (Digitalisation of Rural Areas network)	Demonstrate the application of the model in structuring policy discussions	Researchers	1	Model applied to structure policy discussions and stakeholder input
1st	Demonstration	International focus group (OLLD conference)	ibid.	Researchers and practitioners	1	ibid.
1st	Evaluation	Workshop feedback and reflective discussions	Evaluate the model's ability to support policy development and implementation	Reserachers	1	Methodological insights and identification of limitations (e.g. lack of institutional learning)
2nd	Design & Development	Iterative model refinement based on workshop feedback and literature	Improve the model by addressing identified limitations related to implementation and institutional learning	Researchers		Revised model including a second Double Diamond process focusing on implementation and institutional learning
2nd	Design & Development	Analytical application of the model to literature.	Refine and validate the structure of the model through literature-based analysis	Researchers	1 (42)	Model-informed structuring of literature and refinement of model components
2nd	Demonstration	Policy workshops with researchers, SMALL steering group, and broadband coordinators	Demonstrate the applicability of the revised model in real-world policy development contexts	Researchers and policy stakeholders	5	Model applied to structure policy discussions, stakeholder input, and policy development processes
2nd	Demonstration	Educational application of the model through student projects	Explore the applicability of the model in an educational context	Teachers and students	8	Model applied in 8 student reports
2nd	Evaluation	Feedback from researchers, practitioners, and student applications	Evaluate the model's ability to support both policy development and implementation	Researchers, practitioners, and students		Model supports both needs-driven policy development and implementation processes; indications of improved practical relevance and applicability across contexts
3rd	Design and Development	Educational development of digital platform for ELL through student projects	Understanding of possible implementation from an Information System perspective	Teachers and students	5	Students have made an exploratory conceptual prototype for data sharing in ELL working practices
3rd	Demonstration	Application of the model across five policy areas	Demonstrate the applicability of the model across different policy domains	Researchers		Model applied across domains and comparative insights into its use in different contexts
3rd	Evaluation	Joint research workshop and follow-up analysis across policy areas	Evaluate the model's performance, flexibility, and adaptability across policy domains	Researchers	1	Comparative evaluation across domains; identified strengths and limitations of the model
	Communication	Dissemination through reports, workshops, research funding applications and conferences	Communicate the model, its development, and evaluation results to relevant stakeholders	Policymakers, funders, and practitioners	13	Reports to Swedish national and regional government, reports to funders, conference and workshop presentations, and policy documents based on the model

Table 1: Overview of the DSRM process with three iterations.



Methodological analysis

The policy process was evaluated across the five policy areas based on the empirical material collected during the project. For each policy area, a report was developed synthesising the findings from the needs-driven and institutional processes. Building on these individual reports, a joint evaluation was conducted during a researcher workshop. In this step, all policy areas were analysed collectively using a shared matrix structure. The matrix enabled a systematic comparison across domains by distinguishing between needs-driven and institutional perspectives, allowing us to analyse how understanding developed through participatory co-creation was translated into institutional practice. This facilitated the identification of common patterns, differences, and cross-cutting challenges across the policy areas.

Policy area	Current situation (now)	LLKP*	Desired situation (future)
Needs-driven perspecitve			
Institutional perspective			

Table 2: The methodological analysis matrix. *Living Labs Key Principle.

To explain social phenomena in real-world contexts, where translating practice into policy is inherently challenging, it is necessary to move beyond a focus solely on observable events and instead engage with the underlying processes that shape outcomes (Blom & Morén, 2011; Henfridsson & Bygstad, 2013). This study introduces LL key principles (LL principles) as an analytical lens for examining how understanding of rural needs is developed, translated into policy knowledge, and embedded within institutional practice. In line with critical realism, the LL principles are analysed as generative mechanisms that help explain why specific outcomes emerge in different policy contexts.

(I) Influence: Users and stakeholders are active partners with real influence on the process.
(V) Value: Focus on creating value for all stakeholders. Value is realised in use in real-life contexts.
(R) Realism: Innovation takes place in real-life contexts, understanding of everyday life usage is essential.
(O) Openness: Innovation processes are open to all stakeholders and knowledge sources.
(S) Sustainability: Ensure long-term positive impact by considering ecological, social, and economic aspects, including continuous learning.
(Ståhlbröst, 2012)

Evaluation

The artifact developed in this study is a policy model structured around the DD framework and LL principles, integrating both needs-driven and institutional perspectives into a coherent framework. The first part of the DD frameworks process represents the needs-driven policy process, moving from discovering and defining rural needs and local context to the co-creation of digital solutions and the formulation of policy recommendations (Habibipour et al. 2026). The second process represents the institutional process, focusing on developing an institutional understanding of rural needs and local contexts, defining policy instruments, co-creating institutional solutions, and

supporting implementation and follow-up. The DD/LL framework thus functions as a design artifact that structures both the exploration of needs and the alignment with institutional systems. It supports iterative movement between these two perspectives, reflecting the non-linear and iterative nature of DSRM. Through repeated iterations, the model enables the integration of empirical insights and institutional considerations.

The evaluation across the four policy areas reveals consistent patterns in how needs-driven insights and institutional conditions interact in the co-creation of rural digital policy.

Across all domains, the current situation is characterised by fragmentation, limited cross-border coordination, and a misalignment between everyday needs and institutional structures. From a need-driven perspective, rural conditions such as long distances, uneven access, and limited local resources shape how services are experienced. In education, this is reflected in unequal access to computing and AI learning and limited alignment with children's lived experiences. In health and social care, distance, administrative burden, and system fragmentation constrain accessibility. Environmental governance is marked by underutilised local knowledge and low engagement, while information sharing is limited by low visibility, language barriers, and weak networks.

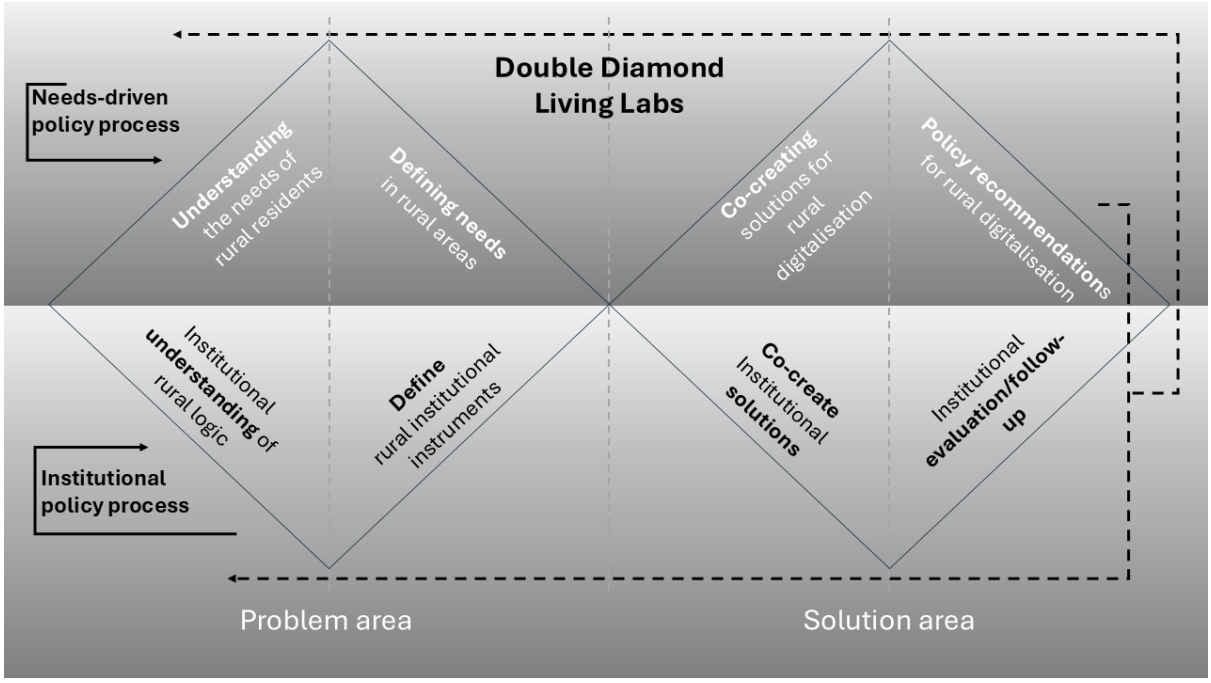


Figure 1: The artifact developed in this study: a policy model structured around the DD framework and LL key principles, interconnecting needs-driven and institutional policy processes across problem and solution areas.

To move from the current state to the desired future state, the LL principles function as underlying generative mechanisms that enable the development of a shared understanding of rural needs, thereby providing the basis for needs-driven policy knowledge. Although these principles are intangible, they shape observable effects within specific contexts. For example, in the current school situation, "children experience hopes and fears regarding digital futures, particularly AI." Through the application of the LL principle of influence, these experiences can be actively



transformed into a desired state in which "education fosters hope, agency, and future orientation". The desired situation across policy areas emphasises accessibility, relevance, and coordination as key qualities of the future state. When understood as processes, these can also be seen as mechanisms through which change is enabled: accessibility facilitates participation, relevance shapes meaning making, and coordination aligns actors across levels and contexts. The principle of value is reflected in how participatory co-creation identifies and prioritises needs that are translated into solutions generating tangible benefits for both residents and public organisations in real-life contexts. This is evident in the emphasis on equal access to services, reduced workload in healthcare, and increased visibility in information sharing, all of which point towards value realised in use. The principle of realism is reflected in how participatory co-creation grounds the understanding of rural needs in everyday practices and lived experiences, ensuring that policy knowledge is developed and validated in real-life contexts. This is evident in the emphasis on equal access to services, reduced workload in healthcare, and increased visibility in information sharing. These findings illustrate how policy knowledge is refined through everyday use; equal access enables engagement with services in practice, reduced workload emerges through actual work situations, and increased visibility develops through real-time information sharing, demonstrating how proposed solutions are continuously validated in real-life contexts.

School	Current situation (now)	LL	Desired situation (future)
Needs-Driven Perspective	•Children experience hopes and fears regarding digital futures, particularly AI •Education does not consistently foster agency or critical understanding •Uneven access due to distance and limited resources •Accessibility linked to participation, not only proximity •Risk of marginalisation •Learning not fully aligned with rural everyday life	I	Education fosters hope, agency, and future orientation
		V	Children develop digital and AI understanding
		R	Education responds to lived experiences
		V	Accessibility, inclusion, equality, safety, environmental and ethical issues addressed
		R	Learning relevant to rural contexts
Institutional Perspective	•Governed by curricula and multi-level actors •Fragmented provision of AI education •Limited cross-border coordination •Institutional constraints and lack of interoperability •Teachers lack support	S	Integrated AI education in curricula
		S	Clear governance and coordination
		O	Strengthened cross-border collaboration
		O	Strengthened remote education
		V	Teacher support and resources
Information sharing	**Current situation (now)**	**LL**	**Desired situation (future)**
Needs-Driven Perspective	•Lack of data information visibility across the border •Language can be a barrier •Lack of network to support information sharing •Local information sharing dependent on very few local stakeholders	V	Cross-border information visibility as driver and enabler for the tourism industry
Institutional Perspective	•Lack of institutional support for or participation in cross-border information visibility	O	Institutional enabling and participation in sharing data to be made visible on the other side of the border



Health & Social	Current situation (now)	LL	Desired situation (future)
Needs-Driven Perspective	•Long distances for patients and healthcare workers •Key aspects: safety, availability, dental care •User needs define care requirements - Limited cross-border cooperation •Costs for cross-border care •High administrative burden and multiple systems •Time-consuming home visits	V	Available and equal care
		S	Digital solutions need to be safe and secure
		I	Needs is the driver for the digital solution
		V	Digital solutions could be used more to free time for workers and increase availability
		I	Have staff on board regarding digital solution / Sense of ownership
		R	Digital solutions can bridge the gap in remote areas, reduce efforts for residents
		R	Digital solutions such as self-tests exist
		O	Possibility for journal cross border
		V	Value in use for digital healthcare solutions
Institutional Perspective	•Data sharing stops at borders (e.g. journals) •Legal and system incompatibility barriers •Limited institutional cross-border cooperation •Systemic barriers (BankID, personal number, analogue processes) •Fragmented and non-interoperable systems	V	Integrated systems to save time for staff
		O	Border not setting the limit for the technology in health care, data management need investigated
		O	Investigate legal and other barriers to digital systems supporting cross-border care
		O	Identify where co-operation over border is necessary
Environment	**Current situation (now)**	**LL**	**Desired situation (future)**
Needs-Driven Perspective	•Local environmental change •Cross-border tensions, regulations and priorities •Rich but informal local knowledge, not integrated into governance •Low engagement due to unrecognised knowledge and agency	I	Local knowledge and lived experience integrated into environmental governance (I)
		S	SDG14,15 recognised as cultural and social value, as well as economic resource
		I	Participatory environmental monitoring with active resident involvement
		O	Strengthened cross-border knowledge sharing and shared responsibility
Institutional Perspective	•Expert-driven, top-down approaches with limited citizen engagement •Weak integration of local knowledge in monitoring •Complex cross-border governance with differing regulations and priorities	S	Integrated participatory environmental governance (ELL)
		I	Monitoring systems informed by local knowledge
		S	Institutional feedback loops to local communities
		O	Strengthened cross-border coordination
		V	Accessible and understandable environmental data for all stakeholders

Table 3: Analysis of current and desired future situations across four domains (school, health and social care, environment, and information sharing/communication), analysed from needs-driven and institutional perspectives.

From an institutional perspective, the principle of openness supports the interpretation, sharing, and integration of needs-driven policy knowledge. The results highlight the importance of openness in enabling integrated systems, clearer governance structures, and strengthened cross-border collaboration. Existing barriers, such as non-interoperable systems, regulatory constraints, and fragmented responsibilities, limit the ability to share knowledge and coordinate across domains. The desired future therefore involves institutional arrangements that support openness through data sharing, coordination, and alignment with local needs. As expressed in the material, this includes "institutional enabling and participation in sharing data to be made visible on the other side of the border," pointing to the need for innovation processes that are open to multiple stakeholders and knowledge sources, thereby enabling institutions to develop a shared understanding of rural needs across organisational and cross-border contexts. The principle of sustainability is reflected in how



need-driven outcomes remain relevant through continuous learning and adaptation. Need-driven outcomes, such as support for agency, participation, and the use of local knowledge in environmental governance, illustrate how sustainability is grounded in long-term relevance and adaptability. These processes enable continuous learning and adjustment over time, ensuring that solutions remain responsive to changing conditions. This is particularly evident in environmental governance, where participatory monitoring and local knowledge create feedback loops that support the development of enduring and adaptive systems. Sustainability is thus generated through mechanisms of needs alignment, participatory knowledge integration, and continuous feedback, which together enable learning, adaptation, and long-term system durability.

Discussion

This discussion returns to the research questions guiding the study. It first considers how Living Lab principles can generate knowledge throughout both needs-driven and institutional policy processes (RQ1) and then examines how a Double Diamond/Living Lab (DD/LL) framework structure and influence institutional learning during policy implementation (RQ2).

LL principles as generative mechanisms for policy development

Within the first, needs-driven DD framework, LL principles provide the participatory mechanisms through which a shared understanding of rural needs is developed through co-creation. Through this process, they shape how local experiences, knowledge, and perspectives are articulated and transformed into a shared basis for policy development. This role can be understood through the concept of generative mechanisms. LL principles can be understood as generative mechanisms in the sense that they operate as underlying causal structures that explain how and why specific outcomes emerge in policy processes. These mechanisms are not directly observable but generate effects through the interaction of actors, contexts, and institutional conditions (Blom & Morén, 2011; Henfridsson & Bygstad, 2013). We have shown that rather than static design principles, LL principles function as drivers of ongoing processes that shape how knowledge is created, shared, and embedded in practice.

Generate understanding of rural needs and local context through participatory co-creation

Together, the LL principles of influence, openness, and realism (Ståhlbröst, 2012) create the conditions for participatory co-creation to emerge as a process through which a shared understanding of rural needs and local contexts is developed. Rather than treating rural needs as predefined, this understanding emerges through dialogue, lived experience, and the integration of multiple stakeholder perspectives. Within the first, needs-driven DD framework process, this shared understanding is articulated, prioritised, and structured into needs-driven policy knowledge, translating diverse experiences and identified needs into coherent policy directions. As a form of needs-driven policy development, this process mobilises both social learning and causal and diagnostic learning (Cerna, 2013).



Structures this understanding into needs-driven knowledge for policy design

While LL principles generate a shared understanding of rural needs through participatory co-creation, the DD framework provides the structured, iterative process through which this understanding is transformed into needs-driven and local context knowledge. Rather than moving directly from identified needs to policy recommendations, the alternating movement between divergent and convergent phases creates space for reflection on diverse experiences, local perspectives, and identified needs, which can then be organised, prioritised, synthesised, and translated into coherent policy directions through successive refinement and stakeholder feedback (Kochanowska et al., 2022; Habibipour et al., 2026). The DD framework was not applied rigidly but adapted to incorporate LL principles, allowing the process to remain responsive to stakeholder input while providing a systematic structure for policy design. This structured policy knowledge forms the starting point for the second interconnected DD framework process, in which public organisations reinterpret and institutionalise the generated knowledge.

LL principles reapplied through the second interconnected DD framework process in the institutional context

While the first DD framework process (Kochanowska et al., 2022) develops a shared understanding of rural needs and local context and structures this into needs-driven policy knowledge, the second interconnected DD framework process reapplies LL principles within institutional contexts. Here, the principles of openness, realism, influence, value, and sustainability (Ståhlbröst, 2012) support public organisations in interpreting, adapting, and integrating needs-driven policy knowledge into their own institutional contexts. Through this process, policy knowledge develops into an institutional understanding of rural needs, providing the basis for institutional learning and context-sensitive policy implementation.

Generate institutional understanding of rural needs and context

The second interconnected DD framework process extends the needs-driven policy process developed by Habibipour et al. (2026) into institutional contexts. Rather than generating new knowledge about rural needs, LL principles are reapplied to enable public organisations to interpret, adapt, and integrate needs-driven policy knowledge within their own institutional settings. In this process, causal and diagnostic learning (Levy, 1994) supports the development of complex place-based understanding by refining actors' understanding of cause-and-effect relationships and linking analytical insights to action-oriented design, while social learning (Hall, 1993) enables mutual understanding among diverse actors involved in the policy process. Together, these learning processes support the development of an institutional understanding of rural needs and contexts, creating the conditions for institutional learning during policy implementation.

Create institutional learning

Institutional learning is created when policy knowledge developed through participatory co-creation becomes embedded within organisational practice. In this case, learning extends beyond



understanding local needs (Habibipour et al., 2026) to the transformation of needs-driven policy knowledge into institutional contexts. Causal and diagnostic learning (Levy, 1994) supports the development of complex place-based understanding by refining actors' understanding of cause-and-effect relationships and linking analytical insights to action-oriented design, while social learning (Hall, 1993) enables mutual understanding among diverse actors involved in the policy process. Institutional learning is created through the interplay of these learning processes during policy implementation. This aligns with the view that mechanisms operate through interacting elements and processes, where outcomes depend on how these are combined in local contexts (Väyrynen et al., 2025). In this way, participatory co-creation supports not only the identification of rural needs, but also their institutionalisation within policymaking organisations.

Co-creation is needed in needs-driven processes to access and construct knowledge about needs and local context, to enable the innovation process, and because needs are not predetermined (Bergvall-Kåreborn & Ståhlbröst, 2008; Galvagno & Dalli, 2014; Ståhlbröst & Holst, 2017). The underlying generative mechanisms for co-creation can be understood as the interaction of the LL principles of influence, openness, and realism (Ståhlbröst, 2012). Influence positions residents and stakeholders as active partners with real influence on the policy process, where their participation and engagement shape the policy areas within the needs-driven process. Openness supports this by ensuring that the policy process is accessible to multiple stakeholders and knowledge sources, thereby facilitating the exchange and integration of diverse perspectives in policy interventions. Realism grounds co-creation in real-life contexts, where policy interventions are based on residents' everyday practices, local conditions, and possibilities for action. Through this interaction, co-creation enables not only the identification of needs, but also the learning processes required for their institutionalisation within policymaking organisations.

To sustain this institutional learning, the sustainability principle emphasises continuous learning and development as central to how LL principles create long-term impact. This involves the capacity of the LL, in this case the policy process, to evolve over time through ongoing knowledge creation and refinement (Ståhlbröst, 2012). In this sense, sustainability operates as a mechanism through which feedback, learning, and adaptation generate enduring outcomes. However, achieving this requires organisations to establish structures that enable trust in the practical judgement of public officials (Bornemark, 2020). Such structures allow knowledge generated through practice to be recognised and embedded within the policy process, rather than remaining informal or marginal. Without this, there is a risk that individuals act outside formal frameworks, as described in street-level bureaucracy (Lipsky, 2010) or in forms of "guerrilla government" (O'Leary, 2019). While learning in these contexts may still emerge through practice and interaction, it remains unevenly integrated and may not contribute to collective or institutional development.

Context-sensitive policy implementation

Context-sensitive policy implementation represents the outcome of the interconnected DD framework, where institutional learning enables policy processes to become responsive to rural contexts and capable of adapting implementation to local conditions. The interconnected DD framework also raises the question of how implementation and institutional learning should be evaluated in rural cross-border policy processes. Conventional key performance indicators (KPIs) are likely to focus on outputs such as the number of workshops conducted, participants involved,



policy documents produced, or digital solutions proposed. While such indicators are relevant, they do not capture the full value of LL-based policy implementation. In the present study, several of the most important outcomes were qualitative and relational, including trust-building, low-barrier participation, the visibility of local knowledge, improved mutual understanding across actors, and the gradual translation of needs into institutional problem framings and solution directions. These aspects are better understood as key value indicators (KVIs), since they reflect whether the process generated socially and institutionally meaningful conditions for implementation (Wikström et al., 2024). In this sense, context-sensitive policy implementation should be assessed not only in terms of performance and delivery, but also in terms of whether policy processes become more responsive to rural realities, more capable of integrating situated knowledge, and better able to sustain coordination and follow-up across organisational and cross-border boundaries. Future research should investigate how key value indicators can be developed to assess whether the LL principles, understood as generative mechanisms, produce the intended outcomes across both interconnected DD framework processes, from the development of shared understanding through participatory co-creation to institutional learning and context-sensitive policy implementation.

This discussion demonstrates how LL principles function as generative mechanisms across two interconnected DD framework processes. Applied first in the rural context, they generate an understanding of rural needs through participatory co-creation, which is subsequently structured into needs-driven knowledge for policy design. Reapplied within the institutional context, the LL principles generate an institutional understanding of rural needs and contexts, creating institutional learning that enables context-sensitive policy implementation.

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